

Segmentation of Lungs and Airways from CT- Scan

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Introduction

Medical image segmentation plays a crucial role in the quantitative analysis of CT scans. It enables the extraction of anatomical structures, measurement of lung volumes, and assessment of abnormalities such as nodules or airway obstructions. In this project, segmentation was applied to a high-resolution CT dataset (CT-CASE15) with the goal of isolating the lungs and airways using two classical techniques: Thresholding and Region Growing. These methods represent fundamental, interpretable, and computationally efficient approaches that form the basis for modern segmentation models such as U-Net and TransUNet.

While numerous advanced methods exist, ranging from clustering and supervised machine learning to deep convolutional and transformer-based neural networks, this report focuses on two classical approaches to provide a conceptual foundation and direct visualization of how segmentation operates on voxel intensity and spatial continuity.

Data Loading and Preprocessing

The CT-CASE15 dataset consisted of 508 axial DICOM slices (512×512 pixels each). Slices were loaded in anatomical order based on their InstanceNumber and stacked into a 3D volume. Raw pixel values were converted to Hounsfield Units (HU) using scanner calibration parameters:

$$\text{HU} = (\text{pixel value}) \times \text{slope} + \text{intercept}$$

This standardization maps air to approximately -1000 HU, lung tissue to -700 HU, and bone to $+1000$ HU, essential for physically meaningful thresholding.

Middle CT Slice

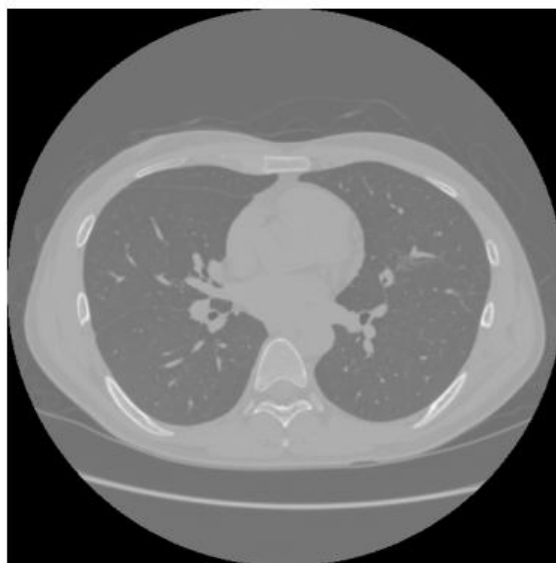
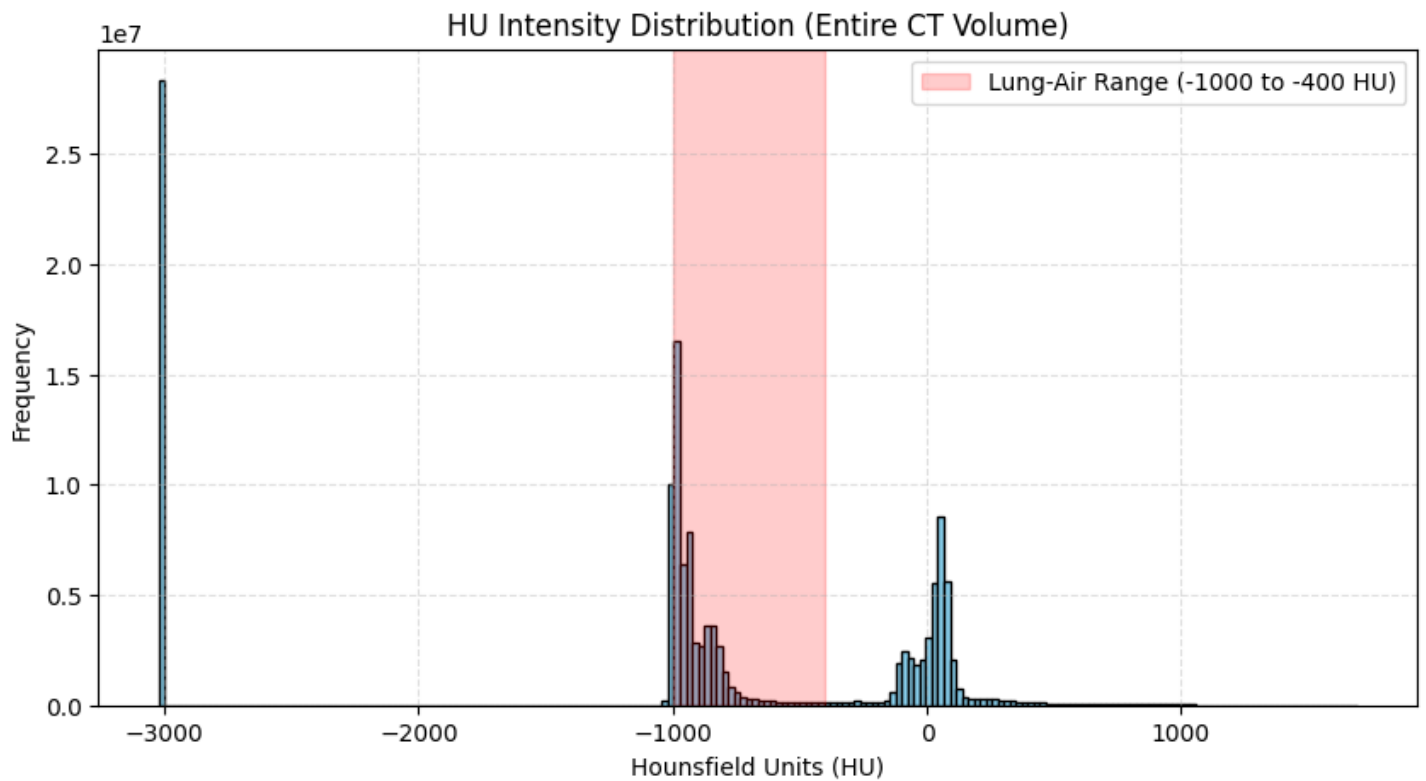


Figure shows a middle axial slice where lungs appear as dark regions (low-density air), while bones and mediastinum appear bright. The histogram reveals three distinct peaks: air (~ -1000 HU), soft tissue (-200 to $+200$ HU), and bone ($> +700$ HU). The range -1000 to -400 HU captures aerated lung parenchyma while excluding denser tissues.



Method 1: Threshold-Based Lung Segmentation

This method uses global intensity thresholds to separate air-filled lungs from surrounding tissues. Each voxel is classified independently based on its HU value, making this a simple, interpretable baseline approach.

The segmentation pipeline began with thresholding, where a binary mask was created by selecting all voxels within the Hounsfield Unit (HU) range of -1000 to -400 , effectively isolating air-filled regions corresponding to the lungs. This was followed by a series of morphological refinement steps to improve segmentation accuracy. A closing operation using a 7×7 kernel was applied to fill small internal gaps within the lung regions, while an opening operation with the same kernel helped remove isolated noise. Subsequently, connected component analysis was performed to retain only the two largest components, representing the left and right lungs. Border clearing was then applied to eliminate any residual external air regions touching the image boundaries.

Results

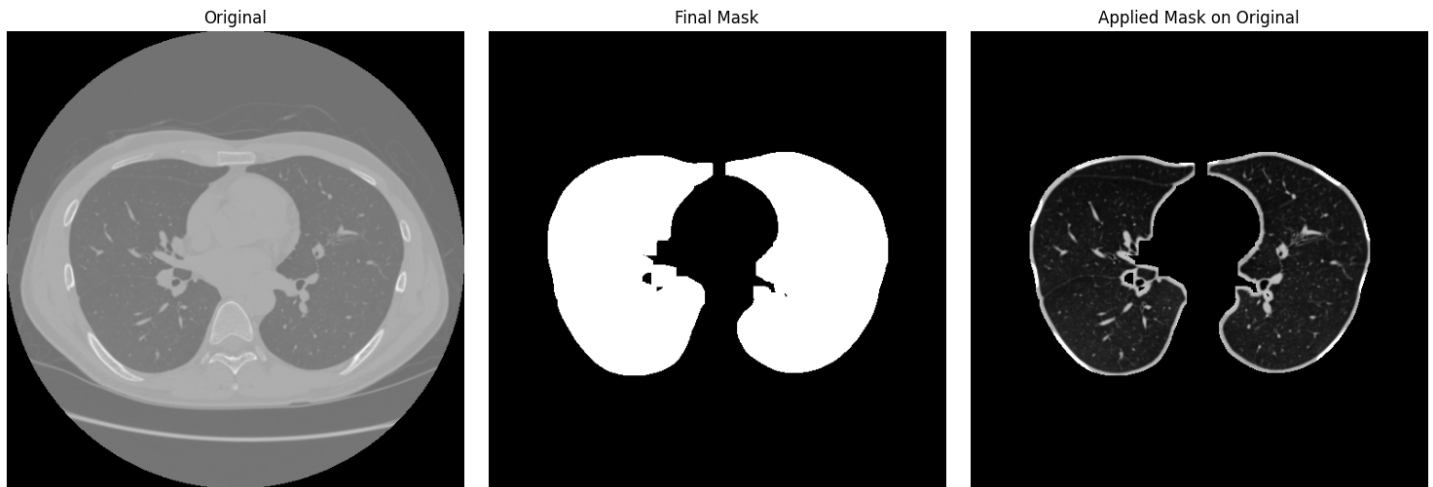


Figure shows the raw threshold mask capturing lung regions but also including external air and internal holes. After morphological refinement, the mask cleanly delineates two continuous lung regions with anatomically consistent boundaries. The overlay confirms accurate parenchyma isolation without significant leakage into surrounding tissues.

Strengths

The threshold-based approach excels in speed and automation, requiring no manual seed points or training data. The entire segmentation pipeline processes the 3D volume in seconds, making it highly efficient for clinical workflows. Each processing step, from HU conversion through morphological refinement which corresponds to well-understood image operations, ensuring transparency and reproducibility. For healthy, non-pathological datasets like CT-CASE15, this method reliably isolates lung parenchyma with anatomically consistent boundaries, providing an excellent baseline for more advanced techniques.

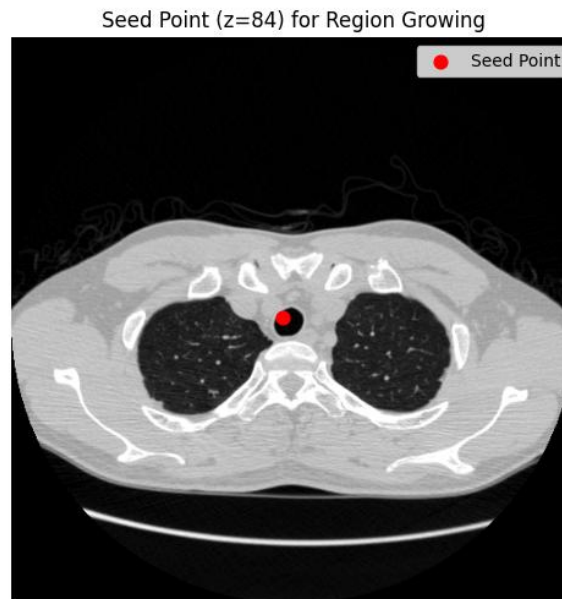
Limitations

Despite its efficiency, this method faces several critical challenges in clinical deployment. Fixed HU thresholds (-1000 to -400) depend on scanner calibration and reconstruction parameters, limiting generalization across different imaging protocols or institutions. The approach fundamentally fails with pathological conditions with consolidations, ground-glass opacities, or dense lesions exhibit higher HU values and are systematically excluded from the lung mask. Leakage to external air remains problematic near the trachea and chest wall, where morphological operations may not completely separate internal from external regions. Hard thresholding inherently produces rough, jagged boundaries that ignore gradual intensity transitions at tissue interfaces. Additionally, morphological kernel sizes require manual tuning for different image resolutions, and the method cannot distinguish internal structures like airways, vessels, or pathological regions within the segmented lung field.

Method 2: Region-Growing Airway Segmentation

Region growing begins from a seed point in the trachea and iteratively expands to neighboring voxels with similar intensities. This connectivity-based method captures anatomically coherent airway structures while excluding unrelated air regions.

Seed Selection and Algorithm



A seed point ($HU \approx -950$) was automatically selected in the tracheal lumen at a superior slice. The 3D region-growing algorithm:

1. Initializes with the seed voxel
2. Examines 6-connected neighbors iteratively
3. Adds neighbors if $-1024 < HU < -400$
4. Constrains growth within body mask to prevent external leakage

Post-processing includes binary closing, opening, and component filtering to smooth boundaries and remove noise.

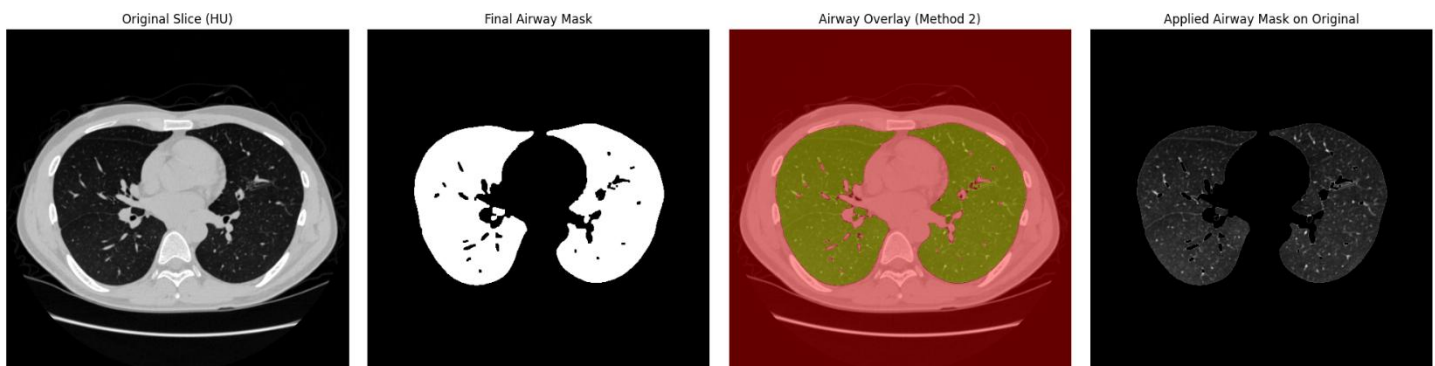
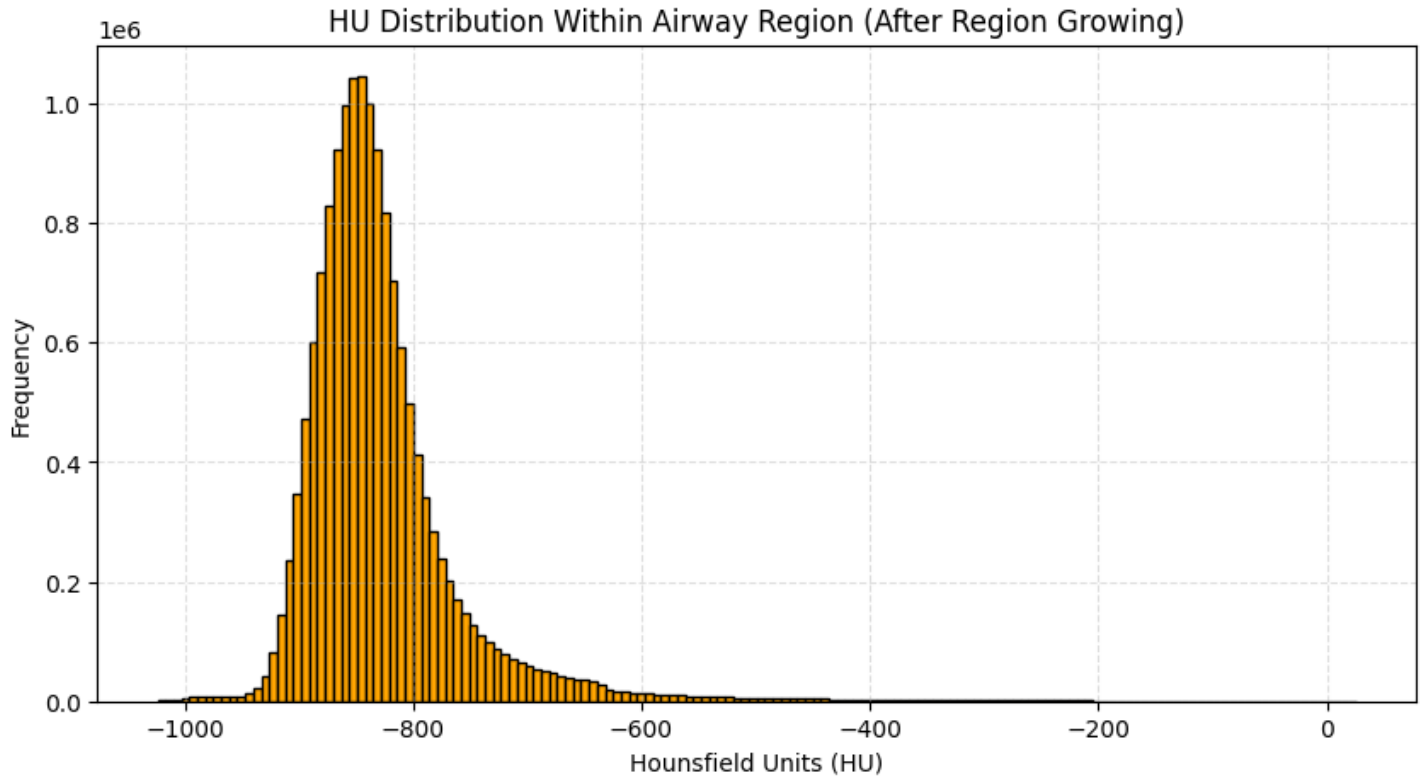


Figure demonstrates the region-grown mask successfully capturing the bronchial tree within both lungs. Unlike thresholding, this produces a spatially coherent, connected structure extending from trachea through major bronchi.



The HU histogram of segmented airways shows a sharp peak at -850 HU (pure air), narrower than the broader lung parenchyma distribution from Method 1. This confirms accurate airway lumen isolation.

Strengths

Region growing provides anatomically coherent segmentation by enforcing spatial connectivity from a single seed point, ensuring the resulting mask represents a genuinely connected airway structure rather than scattered fragments. This connectivity constraint naturally reduces susceptibility to external air inclusion, as the algorithm only expands within the body cavity from the tracheal seed. The method achieves superior boundary precision within the airway lumen, accurately tracing bronchial branches and capturing detailed airway topology that global thresholding cannot distinguish. The resulting segmentation is particularly valuable for anatomical studies, surgical planning, or quantitative airway analysis where structural continuity and branching patterns matter.

Limitations

The method's primary weakness lies in its seed dependency where an incorrectly placed seed can cause incomplete segmentation or expansion into irrelevant regions, though automated seed selection mitigates this in standard cases. The intensity similarity threshold (-1024 to -400 HU) still requires careful tuning for different scanners or patient populations, like global thresholding but now coupled with spatial constraints.

Computational cost increases significantly compared to thresholding, as the iterative voxel-by-voxel expansion process scales with airway complexity and volume size. Potential leakage remains a risk if airways connect to external air through the trachea or boundary discontinuities, requiring careful body mask definition. Partial volume effects at airway boundaries, where voxels contain mixed air and tissue, create irregular mask edges that may need additional smoothing. Finally, in pathological conditions such as mucus plugging, fibrosis, or

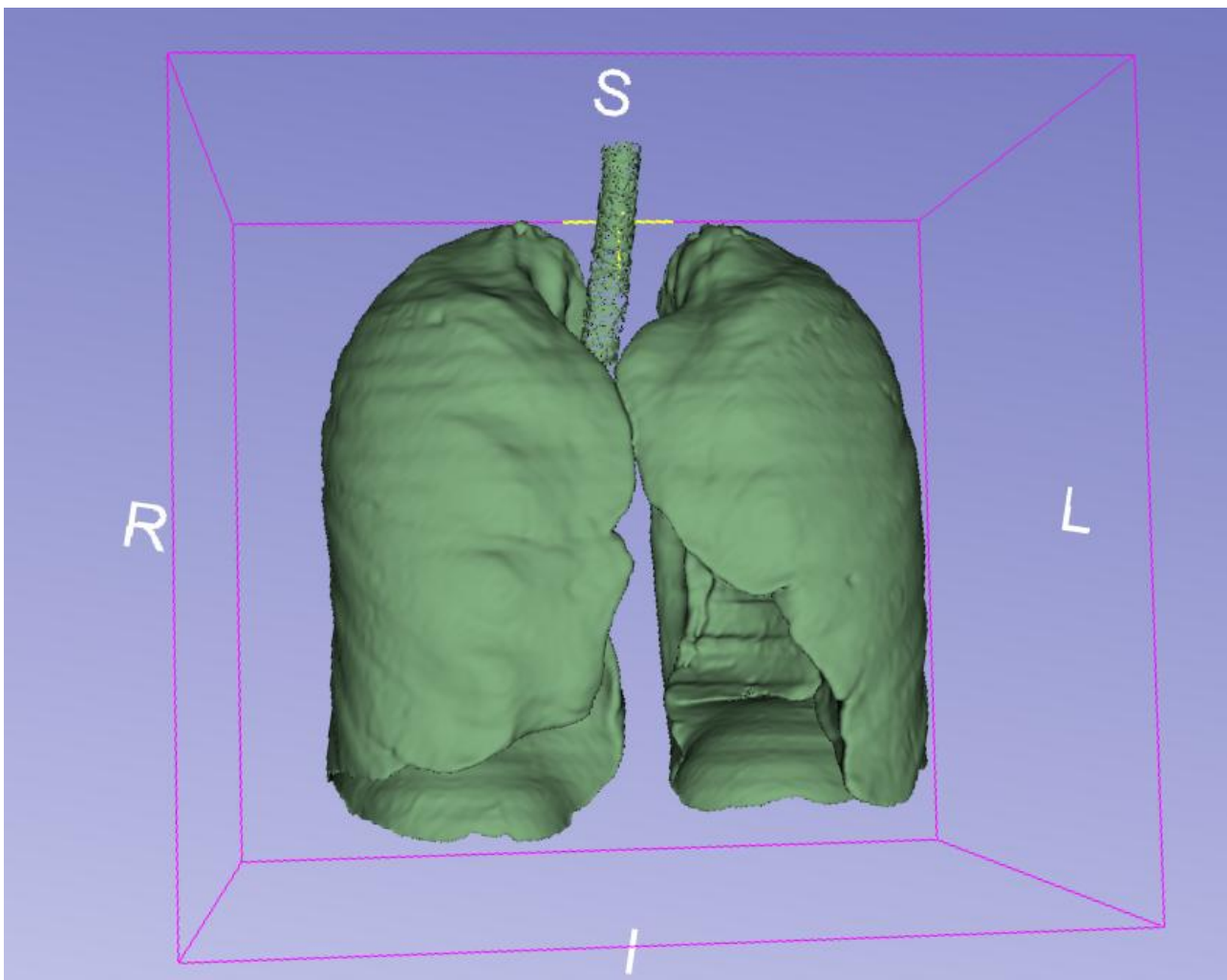
airway collapse, altered intensity patterns can prematurely halt region expansion, resulting in incomplete airway segmentation despite the structural continuity advantage.

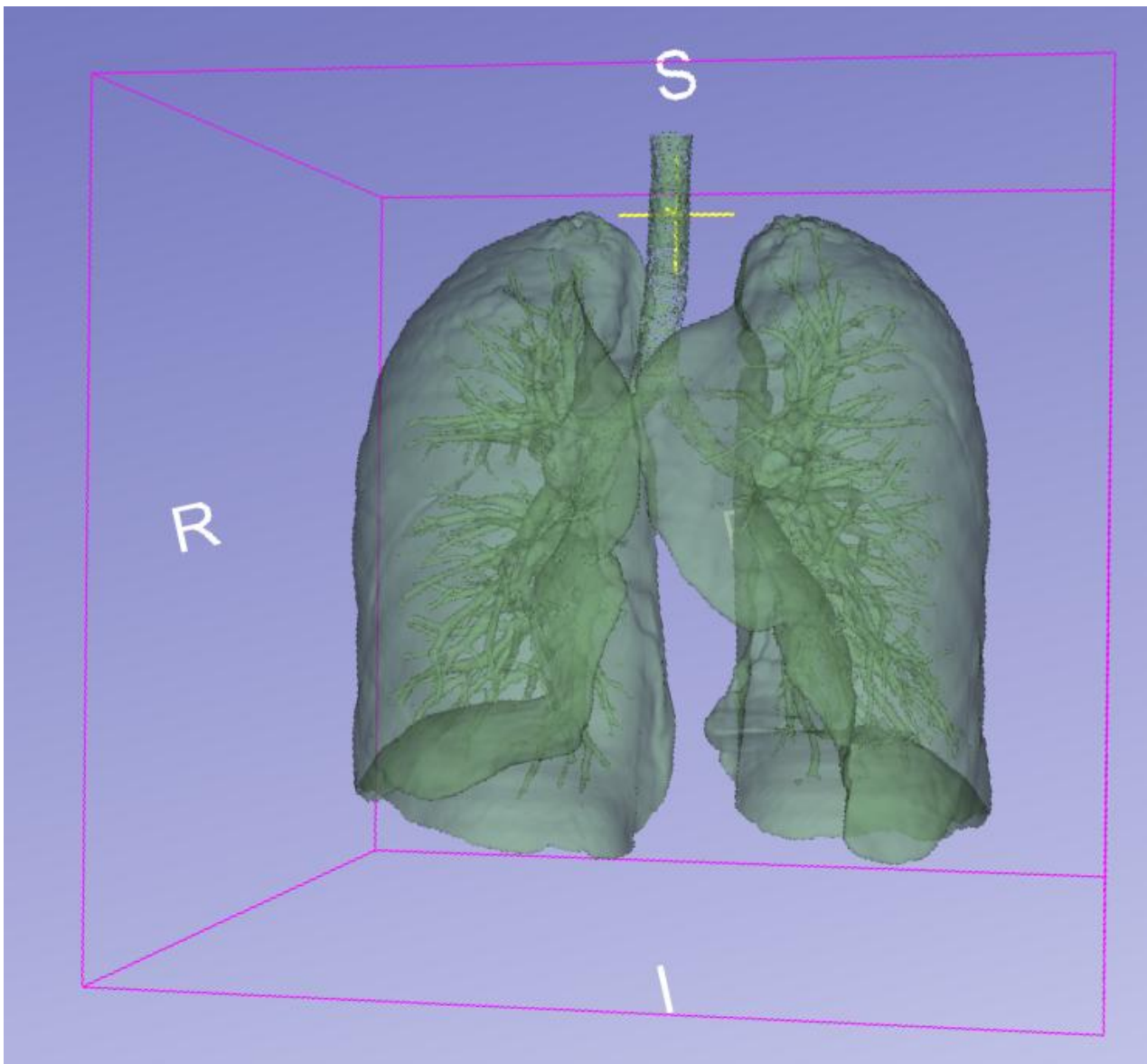
Task II: Lung and Airway Segmentation Using 3D Slicer

The high-resolution CT dataset (CASE15) was imported into 3D Slicer (v5.8.1) for lung and airway segmentation. After loading the DICOM series, the lung parenchyma was segmented using the Threshold tool with intensity values around -600 HU to isolate air-filled regions. Small artifacts and external air pockets were removed using the Islands → Keep largest island function, which yielded a clean binary mask of the lungs.

To extract the airway tree, a second segment labeled “Airways” was created. The Grow from Seeds algorithm was used, where green seeds were placed inside both lungs and red seeds were painted within the trachea and main bronchi. After initialization and refinement, the algorithm successfully propagated airway regions branching within the lungs. Though the lungs and airways appeared in similar color tones, the airways were opaque and dense, while the lungs remained semi-transparent, allowing clear visual distinction in the 3D rendering.

The resulting visualization showed a realistic airway structure embedded inside the lung volume. Minor overlap and small bronchi segmentation limitations were observed due to intensity similarity between adjacent regions. Overall, the segmentation was anatomically accurate and visually consistent, demonstrating effective use of thresholding and region-growing techniques for airway and lung modeling.





Key Insights

The comparative evaluation of thresholding and region-growing segmentation methods highlights their complementary strengths in lung and airway extraction. Thresholding, based on a global intensity cutoff, rapidly isolates air-filled regions across the CT volume using a predefined HU range (-1000 to -400). This approach is computationally efficient and ideal for generating a complete lung field; however, it tends to produce coarse boundaries and may include adjacent low-density tissues or miss pathological regions such as dense lesions and consolidations. Morphological post-processing is therefore essential to refine the binary mask and achieve anatomical consistency.

In contrast, region growing operates through local intensity connectivity, starting from seed points within the trachea or bronchi and expanding iteratively into connected airways that share similar intensity values. This produces smoother, anatomically coherent boundaries and accurately delineates the **airway tree** within the broader lung region. While this method requires more computation and careful seed placement, it demonstrates higher robustness due to spatial constraints that prevent leakage into non-air regions.

Overall, visual and histogram-based analysis confirmed that thresholding captures a broad intensity range encompassing mixed tissues (−1000 to 0 HU), while region growing isolates purer air-filled structures centered near −850 HU. When used together, these techniques form a complementary segmentation pipeline where thresholding provides an initial global lung mask, and region growing refines internal details such as the bronchial branches. This balance between speed and anatomical precision underscores the practical trade-offs inherent in classical medical image segmentation and demonstrates the value of combining multiple approaches for robust and interpretable results.

Conclusion

Both methods produced visually convincing results on the non-pathological CT-CASE15 dataset. Thresholding efficiently segments entire lung volumes and serves as an excellent preprocessing step. Region growing provides superior anatomical detail for airway-specific analysis through connectivity constraints.

These classical techniques form a foundation for understanding medical image segmentation principles. Future work could explore deep learning models (U-Net, TransUNet) that automatically learn intensity, texture, and spatial relationships, enabling robust segmentation across diverse datasets with pathology and varying scanner conditions.

Ps:

Provided the colab notebook in ipynb format and pdf format

References

1. 3D Slicer Tutorial — “*Lung Segmentation using 3D Slicer*”, YouTube. [Available online](#)
2. Lecture code notebooks provided in the course materials (used for thresholding and morphological segmentation reference).
3. ChatGPT (OpenAI, 2025) — used to generate visualizations and phrasing refinement

